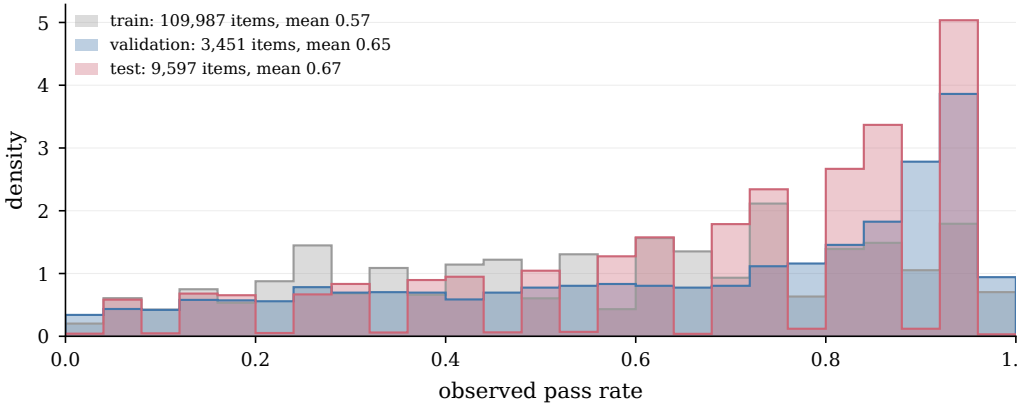
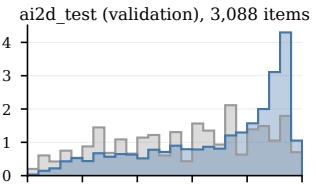
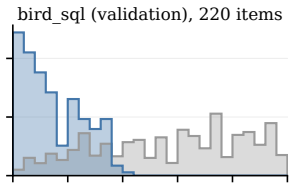
Observed item difficulty by split

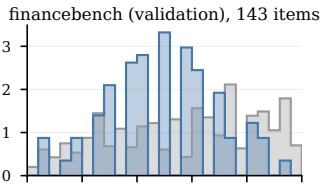
Pass rate = share of subjects answering the item correctly; items with ≥5 responses. Lower means harder.

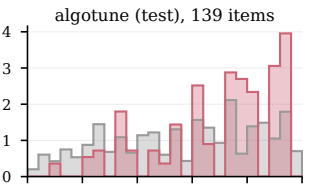


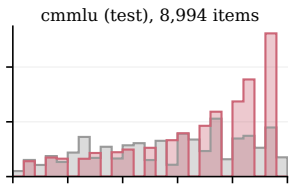
Each held-out benchmark against the pooled training distribution (grey).

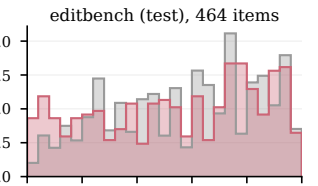












Does the model's item belief track observed difficulty?

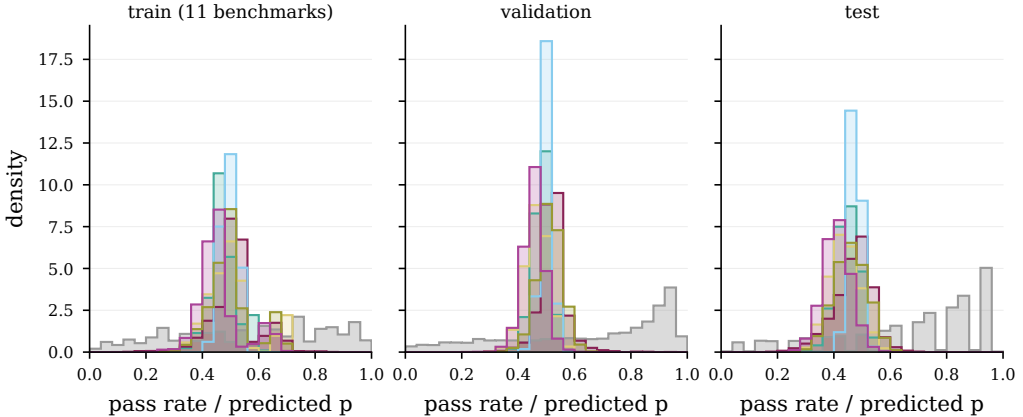
Budget-0 prediction (empty context) vs observed pass rate, per item. r Pearson, ρ Spearman, sd = spread.

model	split	items	r	ρ	mean pred	mean obs	sd pred	sd obs
ANP	train	109,987	+0.186	+0.172	0.475	0.570	0.050	0.267
ANP	validation	3,451	+0.065	+0.050	0.483	0.651	0.031	0.280
ANP	test	9,597	-0.050	-0.053	0.448	0.673	0.041	0.253
ANP, budget-ladder training	train	109,987	+0.124	+0.075	0.498	0.570	0.088	0.267
ANP, budget-ladder training	validation	3,451	-0.060	-0.070	0.466	0.651	0.044	0.280
ANP, budget-ladder training	test	9,597	-0.061	-0.060	0.436	0.673	0.053	0.253
ANP + SAM	train	109,987	+0.178	+0.160	0.516	0.570	0.081	0.267
ANP + SAM	validation	3,451	+0.027	+0.038	0.522	0.651	0.050	0.280
ANP + SAM	test	9,597	-0.079	-0.083	0.467	0.673	0.069	0.253
ANP + prior anchor	train	109,987	+0.084	+0.050	0.497	0.570	0.032	0.267
ANP + prior anchor	validation	3,451	+0.140	+0.069	0.499	0.651	0.018	0.280
ANP + prior anchor	test	9,597	-0.051	-0.059	0.474	0.673	0.021	0.253
ANP + SAM, budget-ladder training	train	109,987	+0.185	+0.163	0.501	0.570	0.079	0.267
ANP + SAM, budget-ladder training	validation	3,451	+0.032	+0.035	0.511	0.651	0.044	0.280
ANP + SAM, budget-ladder training	test	9,597	-0.083	-0.082	0.460	0.673	0.058	0.253
ANP, dense-prefix training	train	109,987	+0.191	+0.162	0.461	0.570	0.078	0.267
ANP, dense-prefix training	validation	3,451	+0.071	+0.058	0.455	0.651	0.038	0.280
ANP, dense-prefix training	test	9,597	-0.053	-0.055	0.410	0.673	0.050	0.253
ANP, random-subject split	*train*	*109,987*	*+0.310*	*+0.303*	*0.555*	*0.570*	*0.157*	*0.267*
ANP, random-subject split	*validation*	*3,451*	*+0.606*	*+0.640*	*0.632*	*0.651*	*0.187*	*0.280*
ANP, random-subject split	*test*	*9,597*	*+0.039*	*+0.023*	*0.546*	*0.673*	*0.164*	*0.253*
ANP, random-subject split (width 12	*train*	*109,987*	*+0.271*	*+0.284*	*0.467*	*0.570*	*0.114*	*0.267*
ANP, random-subject split (width 12	*validation*	*3,451*	*+0.604*	*+0.610*	*0.628*	*0.651*	*0.118*	*0.280*
ANP, random-subject split (width 12	*test*	*9,597*	*-0.009*	*-0.010*	*0.487*	*0.673*	*0.108*	*0.253*

A model that learned transferable difficulty would keep r and its spread on validation and test; one that memorised its training benchmarks keeps them only on train. Italic rows: random-subject split, which also trained on ai2d_test and bird_sql (validation) and editbench (test); see the seen-vs-unseen page.

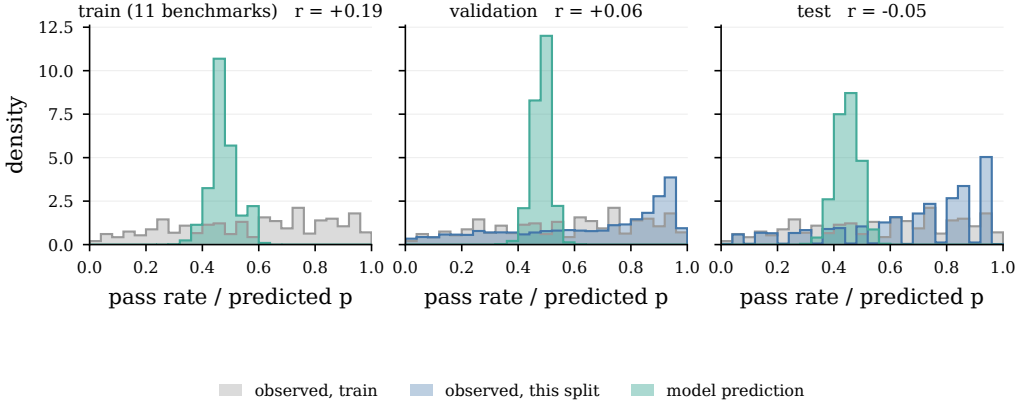
All holdout models' item beliefs, by split

Grey: observed pass rate on that split. Colours: each model's budget-0 prediction, filled at low opacity.



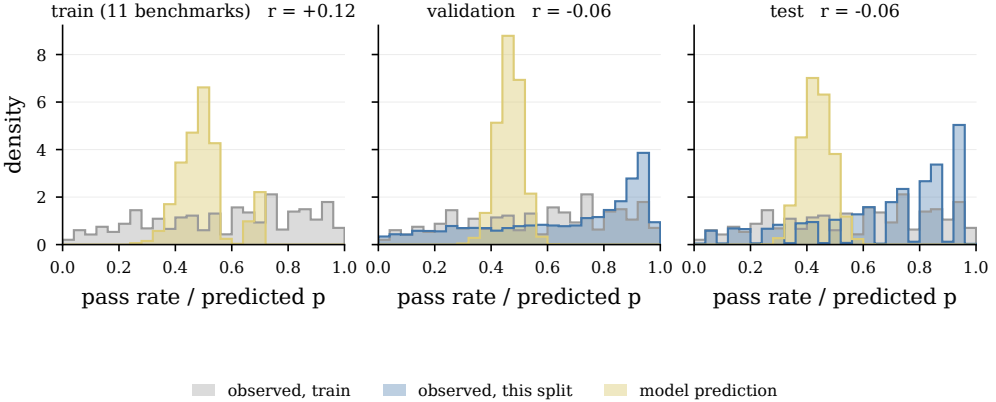
ANP: predicted vs observed item difficulty

Grey: train obs. Blue: this split obs. Colour: budget-0 prediction.



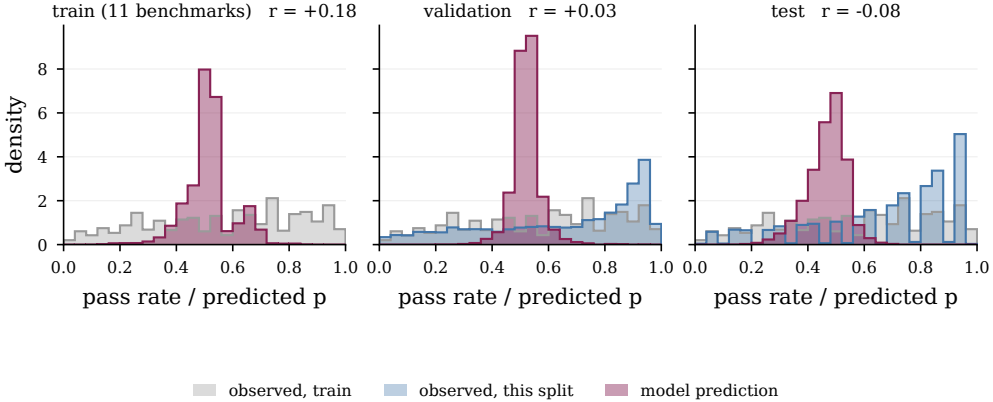
ANP, budget-ladder training: predicted vs observed item difficulty

Grey: train obs. Blue: this split obs. Colour: budget-0 prediction.



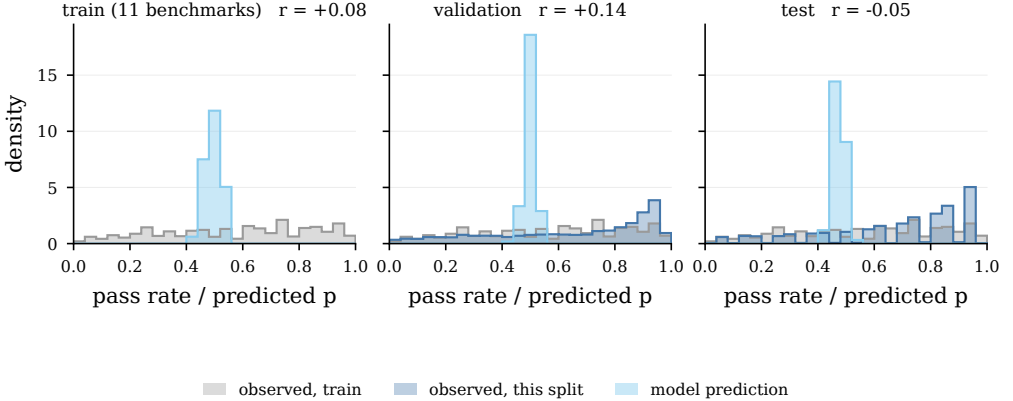
ANP + SAM: predicted vs observed item difficulty

Grey: train obs. Blue: this split obs. Colour: budget-0 prediction.



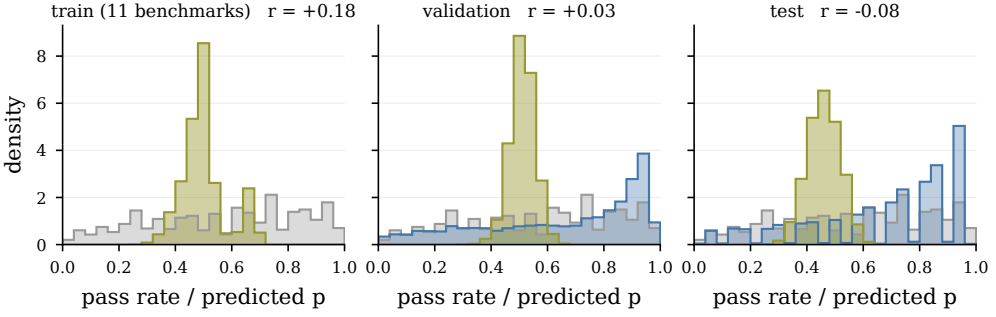
ANP + prior anchor: predicted vs observed item difficulty

Grey: train obs. Blue: this split obs. Colour: budget-0 prediction.



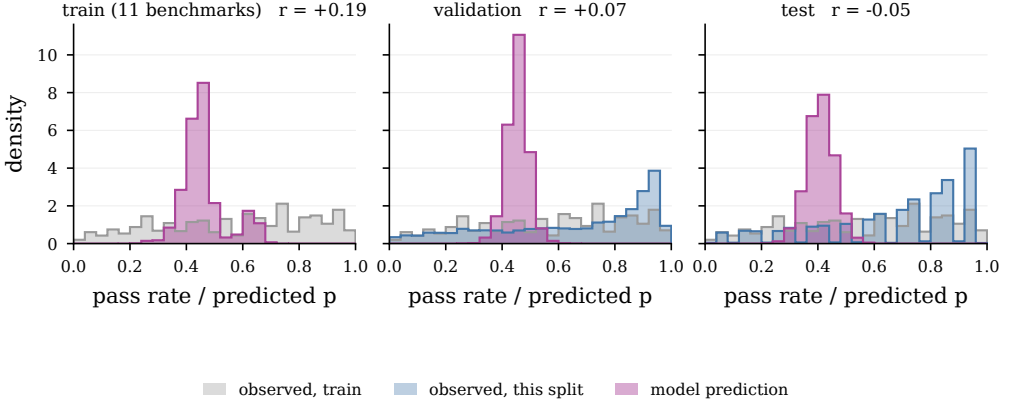
ANP + SAM, budget-ladder training: predicted vs observed item difficulty

Grey: train obs. Blue: this split obs. Colour: budget-0 prediction.



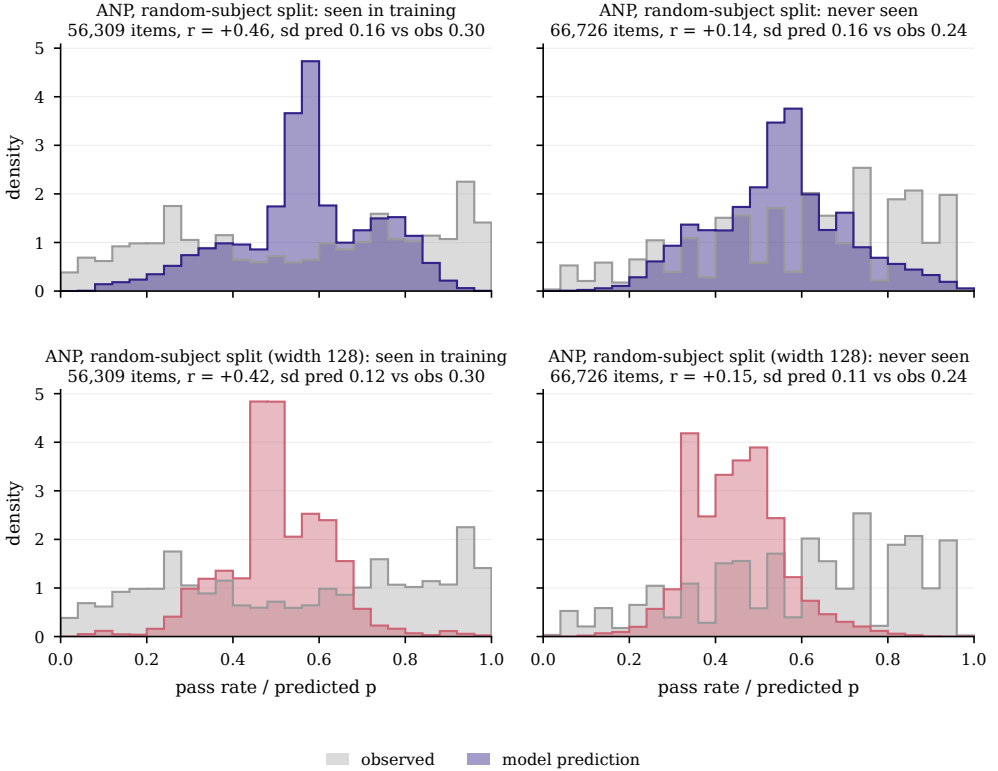
ANP, dense-prefix training: predicted vs observed item difficulty

Grey: train obs. Blue: this split obs. Colour: budget-0 prediction.



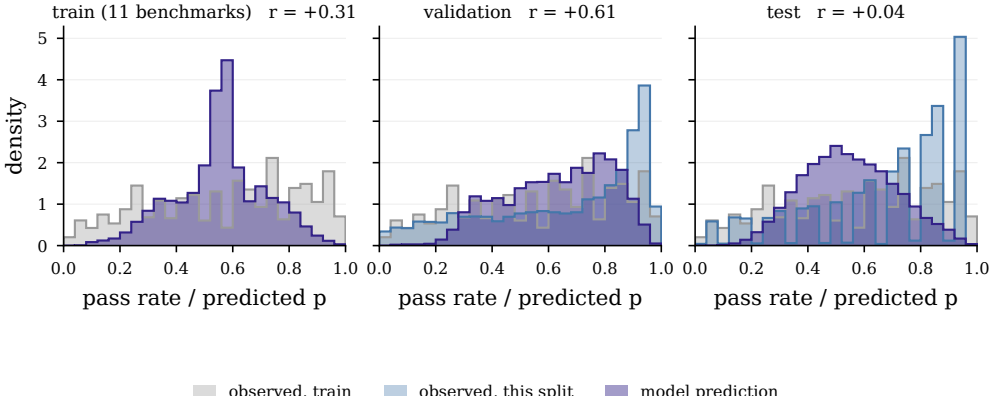
Random-split models: benchmarks seen in training vs never seen

Grey: observed. Colour: budget-0 prediction. Seen = benchmark in the 7 trained on.



ANP, random-subject split: predicted vs observed item difficulty

Random-subject split: trained on 7 benchmarks incl. ai2d_test, bird_sql, editbench. Grey: train obs. Blue: this split obs. Co



ANP, random-subject split (width 128): predicted vs observed item difficulty

Random-subject split: trained on 7 benchmarks incl. ai2d_test, bird_sql, editbench. Grey: train obs. Blue: this split obs. Co

